

Behavioral/cognitive description of autism: some “modeling theories”

Executive functions

- deficit in executive functions
 - concept formation, cognitive flexibility, fluency, action planning, response inhibition, and working memory
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Focus on details / integration / complexity

- weak central coherence (WCC): cognitive style focused on details → reduced global processing or contextual integration
 - implicit language: requires integration of multiple information sources + context
 - social: difficulty understanding social situations globally (over-focus on specific elements, e.g., one word)
 - limited interests / attraction to regularities
 - initial version: inability to integrate information
 - later version: local processing bias rather than a true deficit
- Baron-Cohen (2006): hyper-systemizing theory of autism
 - systemizing mechanism set too high in autistic individuals
 - difficulty coping with high-variance systems
 - parallels with WCC but framed as adaptive: detail focus serves system understanding
- monotropism (Murray, 2005): narrow range of interests // related to WCC
 - competition for attention shapes cognitive processes
 - lack of generalization
 - * top-down processing: strict / limited / resistant to correction
 - * hyper-awareness within attentional focus (“attentional tunnel”), hypo-awareness outside
- Dwyer (2024): supports monotropism account of ASD
 - strong sustained attention on one subject, difficulty dividing attention
 - atypical regulation of attention rather than a simple deficit
 - ASD: both
 - * increased endogenous hyper-focus
 - * increased susceptibility to exogenous attentional capture (inattention)
 - greater hyper-focus correlated with lower quality of life (QOL)
 - * but positive effects on QOL mediated by meaningful engagement in the focused activity
- Ben-Yosef (2016)
 - compares three accounts:
 - * weak central coherence (local vs global)
 - * complexity account
 - * social cognition deficit
 - no deficit in processing human stimuli, nor in global processing (conceptual or perceptual)
 - * despite previous findings reporting deficits
 - intact context processing in direct associative contexts (e.g., image + sound of a dog)
 - impaired integrative context processing (no single cue directly leads to the target; requires integration)
 - conclusion: difficulty when synchronization between brain regions is required, but not limited to this
 - * also general cognitive overload

- * supports complexity account of ASD
 - Mottron (2000)
 - cognitive differences observed across domains (e.g., visuospatial, music, language, memory) in ASD
 - * hypothesis: multi-modal deficit?
 - deficit across sensory modalities or across cognitive operations within a modality?
 - * alternative (WCC): deficit in integration into higher-level organization → detail focus
 - hierarchization deficit hypothesis (HDH):
 - * local and global processing intact
 - * but global processing has no advantage over local processing
 - same/different judgments with:
 - * local change (note change)
 - * global change (transposition) in melodies
 - findings:
 - * no impairment in global processing
 - * but local processing advantage
 - interpretation:
 - * enhanced perception of elementary sound properties rather than integration deficit ?
 - Paradoxical Functional Facilitation:
 - * neurological deficit may induce enhanced performance
 - * preserved perceptive processes compensate for impaired non-perceptive ones
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Atypical sensory processing

- overreactivity as a primary symptom in autism (Liss, 2006)
 - hypothesis: individuals with ASD experience over-arousal → leads to over-selective attention and repetitive behaviors
 - cluster analysis: subgroup with both overreactivity and over-selective attention
 - * over-focusing group: highest cognitive functioning but most socially impaired
 - over-focus detrimental to social behavior but not to intellectual functioning?
 - * lowest-functioning group: more under-reactive
 - possibly linked to intellectual disability
 - hypothesis: over-selective attention and repetitive behaviors as compensation mechanisms
 - * predictable/repetitive events regulate arousal levels
 - over-arousal hypothesis reformulated as “noisy neuronal firing”
 - * Hussman (2001): reduced inhibition (GABAergic neurons)
 - * Casanova (2002): cortical minicolumns smaller and more numerous → overconnectivity, reduced selectivity
 - * Kinsbourne: exaggerated ascending activation from brainstem to cortex
- Gomes (2008): hypersensitivity
 - sensory abnormalities linked to attention
 - high reactivity associated with increased attention, and vice versa
- Hilton (2007): link between sensory and social profiles
 - Dunn (1997) sensory processing model:
 - * low/high threshold × active/passive response
 - moderate to strong relationship between sensory responses and social functioning

- strongest for:
 - * sensation avoiding (low threshold, active): avoidance of overwhelming social interactions?
 - * sensory sensitivity (low threshold, passive): distractibility / hyperactivity → missed social cues
 - moderate for:
 - * sensation seeking (high threshold, active): atypical behaviors may discourage peers
 - * low registration (high threshold, passive): reduced detection of sensory input → missed social cues
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Empathy

- Milton (2012): double empathy problem
 - non-autistic individuals may also lack understanding/empathy toward autistic individuals
- Crompton (2020): “telephone game” paradigm
 - autistic–autistic and non-autistic–non-autistic chains: effective communication
 - mixed chains: reduced efficiency
 - interpretation:
 - * communication difficulties arise from neurotype mismatch, not solely from deficits
 - * information transfer may be more effective within same neurotype
- Alkhalidi (2021):
 - neurotypical individuals report disliking autistic individuals more
 - perceived as awkward and lacking empathy
- Dunn (2023):
 - autistic individuals rate themselves as awkward
 - but still show willingness to engage socially
 - suggests they are not universally perceived as “undesirable”

Other

- Female Autism Phenotype (FAP) (Hull, 2020)
 - higher social motivation, but difficulty maintaining long-term relationships
 - camouflaging: use of conscious or unconscious strategies (explicitly learned or implicitly developed) to minimize the appearance of autistic characteristics during social situations
 - * more difficult to diagnose, often misdiagnosed as anxiety, etc.
 - * strong associations between camouflaging and poor mental health outcomes
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Findings from cognitive tests / behavioral measures

- Lau (2012): high prevalence of ASD among AgCC, but ASD traits also present in many parents, AgCC not
 - AgCC shows intermediate scores between control and ASD
 - * suggests only part of the explanation?
- Wheelwright (2010): Autism Spectrum Quotient (AQ)
 - NAP: narrow autism phenotype (>3 SD)
 - BAP: broad autism phenotype (>1 SD)
 - MAP: medium autism phenotype (>2 SD)
 - higher prevalence:
 - * males > females

- * scientists, especially mathematicians
 - * parents of autistic individuals
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Findings about neural correlates in ASD

- Piven (1997): smaller corpus callosum (CC), larger cortical regions (especially parietal)
 - CC smaller only when controlling for brain size
 - * may reflect enlarged cortical regions rather than true CC reduction
 - * especially in regions without strong CC connections
- Just (2006): fronto-parietal underconnectivity
 - may arise from anatomical constraints (e.g., smaller genu of CC)
 - Casanova (2002): more short-range fibers than long-range → deficit in long-distance connectivity
 - executive dysfunction possibly linked to fronto-parietal underconnectivity
 - normal activation within individual regions
 - * difficulties arise in tasks requiring integration across regions
 - theory of mind, face processing, executive functions → all rely on integration
 - postural deficits? (Minshew, 2004)
 - conclusion:
 - * interest in studying functional connectivity in individuals with complete or partial agenesis of the corpus callosum (AgCC) with preserved cognition
 - * to understand how absence of CC affects interhemispheric coordination
- Mason (2008):
 - three types of inference:
 - * intentional (theory of mind-based)
 - * physical
 - * emotional
 - ASD: increased right hemisphere (RH) activation during reading for all inference types and context sentences
 - * hypothesis: spillover effect → RH recruited due to left hemisphere (LH) overload
 - reduced functional connectivity:
 - * within theory of mind network (left medial frontal gyrus <-> right temporo-parietal junction)
 - * between theory of mind network and LH language network (left inferior frontal gyrus, left superior temporal gyrus)
 - functional connectivity (frontal–parietal) correlated with size of CC genu
- Alexander (2007):
 - only a subgroup of autistic individuals shows decreased fractional anisotropy (FA) / increased mean diffusivity
 - * driven by increased radial diffusivity
 - suggests myelination deficit
 - correlated with CC size and processing speed
 - similar findings in Shukla (2010)
- Belmonte (2004):
 - ASD: high local connectivity + low long-range connectivity (cf. Just et al., 2004)
 - overconnectivity in sensory areas:
 - * excessive activation (low selectivity)

- * no clear distinction between target and non-target stimuli
 - * leads to overload at later processing stages
 - * integrative areas receive degraded input → reduced activation
 - Belmonte (2004): cerebellum
 - implicated in long-range connectivity and coordination of cognitive functions
 - ASD: smaller cerebellum, fewer Purkinje cells (inhibitory)
 - * high physical connectivity but low computational connectivity
 - * linked to cognitive, motor, and social deficits
 - Rubenstein (2003): modulated noise exposure model of hypersensitivity
 - critical developmental window for establishing excitation/inhibition balance and sensory maps
 - * also influences closure of critical periods in cortex
 - modulated noise: early closure → poor differentiation (low signal-to-noise ratio)
 - continuous noise: no closure → absence of stabilization signal
 - GABAergic neurons:
 - * lateral inhibition (between cortical columns)
 - * intra-column regulation (prevent excessive activation)
 - * timing of synaptic plasticity
 - Dapretto (2005): mirror neuron system (MNS)
 - no MNS activation during observation/imitation of emotional expressions in ASD
 - behavioral performance similar to controls (no difference in fixation duration on eyes/faces)
 - suggests reliance on visual/motor attention rather than MNS
 - MNS activity correlated with social ability
 - Diamond (2019): voxel-based analysis in ASD
 - decreased fiber density (FD) in CC (splenium and genu) → reduced axonal density?
 - decreased global white matter FD (GWM-FD), but smaller effect than in CC
 - * suggests more widespread (but milder) alterations
 - negative correlation between social impairment and fiber density in CC splenium
 - Paul (2012) hypothesis:
 - when the brain lacks callosal connections (e.g., AgCC), it develops alternative pathways
 - * to maintain bilateral resting-state networks (RSNs) and interhemispheric connectivity
 - callosotomy data suggest:
 - * minor commissures and thalamocortical pathways may support compensation
 - Demopoulos (2015): conceptual model of “sensory processing inefficiency”
 - hypothesis: impaired interhemispheric communication via CC contributes to inefficient information processing
 - * difficulty integrating relevant social information
 - CC may support language learning and cortical development by:
 - * enabling rapid transfer of auditory and visual information
 - * toward dominant language hemisphere for integration and interpretation
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General model for hypotheses on autism (inspired by Rubenstein, 2003)

- 1. Early genetic or environmental alteration (gene mutation or atypical environment)

- 2. Reduced number of GABAergic neurons in cortex/amygdala (reduced production or migration)
- 3. Reduced inhibition (GABA) relative to excitation (glutamate) → excitation/inhibition imbalance
- 4. Excessive plasticity (e.g., increased long-term potentiation, LTP)
 - overly rapid learning, reduced flexibility
 - increased reactivity to stimuli
- 5. Underdifferentiated representations
 - simple stimuli activate large neuronal populations
- 6. Local hyperexcitability → reduced differentiation → decreased long-range excitability
 - leads to reduced long-distance connectivity
- 7. Emergence of atypical behaviors (e.g., hypersensitivity, rigid memory, reduced flexibility, increased epilepsy risk)

Neural correlates of AgCC

- anatomy (Tovar-Moll, 2006):
 - some properties preserved, others strongly altered → compensatory or maladaptive?
 - partial AgCC (pAgCC):
 - * hypothesis: genu (anterior CC) forms first, then progressively more posterior regions
 - consistent with remnant CC mainly consisting of genu
 - * preserved topographic organization similar to typical CC
 - remnant genu connects mainly frontal regions (no posterior expansion)
 - general organization: frontal/genu, parietal/body, occipital/splenium
 - Probst bundles (PB): ipsilateral bundles, not seen in controls or contralateral hemisphere → not an artifact
 - * present in majority of cases + preserved topographic organization → compensatory?
 - sigmoid bundle (SB): S-shaped, connects right frontal lobe to left occipito-parietal regions
 - * present in ~30% of patients, often those most severely impaired → maladaptive?
 - Lazarev (2016): PB and SB may support synchrony between heterotopic regions (EEG coherence)
- functional connectivity (synchronization of signals)
 - Hinkley (2012): reduced connectivity in some cortical regions (resting-state alpha)
 - * frontal, parietal, occipital regions → linked to altered cognitive functions in AgCC
 - * activity still present, but reduced connectivity
 - * correlated with executive function measures
- structural connectivity (physical pathways)
 - Shi (2021):
 - * reduced interhemispheric connectivity
 - * increased intrahemispheric connectivity
 - * structural alterations not strongly associated with broad neurobehavioral outcomes in children with AgCC
- Owen (2013):
 - qualitative organization of resting-state networks similar between controls and AgCC
 - reduced interhemispheric connectivity in associative networks, not sensory networks
 - * thalamus may maintain symmetric bilateral connectivity (especially for sensory processing)

- Hett (2006): anterior commissure ~10% larger in AgCC
 - * correlated with better cognition (Barr, 2002)
 - high variability across AgCC individuals
 - * multiple compensatory strategies → difficult to study
 - Yuan (2020):
 - link between structural connectivity (SC) and functional connectivity (FC)
 - * faster-conducting fibers → stronger synchronization of neural signals
 - increased intrahemispheric SC/FC → possible compensatory mechanism in AgCC
 - compensation hypothesis: increased lateralization to compensate for reduced interhemispheric connectivity
 - * more left-handed in (small) population AgCC and right lateralization of left limbic area
 - Szczupak (2020):
 - AgCC alters interhemispheric synchrony of cortical communication
 - follows developmental gradient:
 - * early-developing regions (antero-posterior, medio-lateral CC development) less affected
 - posterior sensory regions (homotopic connections) more affected
 - cingulate cortex (pioneer axons): preserved synchrony but delayed
 - myelinated CC axons connect homotopic regions and terminate in cortex
 - modulate activation/inhibition (GABAergic neurons)
 - regulate excitation/inhibition balance
 - enable interhemispheric inhibition → control lateralization and synchronization
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Subregions

- 7 subregions (Witelson method)
 - in some studies: only anterior, middle, posterior divisions
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Genetics

- Tick (2015): meta-analysis of twin studies in ASD
 - monozygotic twins (MZ): correlation // 0.98
 - dizygotic twins (DZ): correlation // 0.53
 - heritability estimates: 64–91%
 - extreme cases do not show higher heritability → supports continuum rather than distinct categories
- Lau (2012):
 - in AgCC: structural abnormalities often associated with de novo chromosomal mutations
 - low recurrence risk in families → behavioral phenotypes may also be discontinuous (Sherr et al., 2005)
- Lau (2012): shared ASD–AgCC genetic factors
 - ARX gene mutations (affect connectivity and neuronal excitation)
 - Schell (2003): ARX (Aristaless Related Homeobox)
 - * critical for brain and neuronal development
 - * associated with widespread central nervous system malformations
 - * linked to autism, AgCC, dystonia, epilepsy, thalamic dysplasia, West syndrome
 - Schell (2005):

- * often de novo mutations
 - * insertions/deletions identified via CGH array (comparative genomic hybridization)
 - * males more severely affected (single X chromosome)
 - * phenotype depends on gene dosage (e.g., deletion = 1 copy, duplication = 3 copies)
 - Belmonte (2004):
 - CMH protein involved in synaptic pruning altered in ASD
 - cerebellar alterations (Purkinje cells) → social deficits
 - Rodriguez (2021): three gene categories involved in ASD
 - neurotransmitter receptors (e.g., serotonin, dopamine, GABA)
 - genes involved in neurotransmitter metabolism (e.g., tryptophan → serotonin, melatonin)
 - cell adhesion molecules and scaffolding proteins
 - Wahlsten (2003): mouse model of AgCC
 - BTBR T/+ tf/tf strain: AgCC + reduced hippocampal commissure
 - * cross between heterozygous T gene and homozygous tufted mutation
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Gender differences

- ASD more prevalent in males
 - unclear whether due to diagnostic bias or true biological differences
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Role of the corpus callosum (CC)

- Banich (1998):
 - CC enables transfer of sensory information and coordination between hemispheres
 - role in attention: allows division of labor between hemispheres → increased computational capacity
 - * dynamic interhemispheric interaction increases processing capacity
 - * trade-off between computational gain and integration cost
 - two hypotheses:
 - * “gating” mechanism when one hemisphere exceeds capacity
 - * CC as an attentional control system
 - attention conceptualized as:
 - * resource
 - * selective gating mechanism
- Yuan (2020):
 - increased intrahemispheric connectivity → compensatory in AgCC
 - possible increased lateralization
- Hinkley (2016):
 - pAgCC: bilateral language representation
 - complete AgCC (cAgCC): right-lateralized language
 - reduced left lateralization linked to lower verbal IQ (VIQ) → clinical implications
 - some non-language regions less activated in LH
 - * compensation may come at cost of reduced processing in these regions

- supports CC role: interhemispheric inhibition (suppress one hemisphere while the other is active)
 - Karolis (2019):
 - interhemispheric independence hypothesis
 - * brain expansion → functional lateralization to reduce conduction delays
 - * lateralized regions less strongly connected via CC
 - improves efficiency but reduces recovery after lesions
 - Gazzaniga (2000): evolutionary perspective
 - lateralization = development of new skills, maintenance of others
 - new functions (e.g., language) emerge in one hemisphere
 - other hemisphere maintains older functions
 - * lateralization → specialization + preservation
 - non-lateralized processing allows parallel processing (requires CC coordination)
 - CC enables specialization and coordination
 - * LH: language, problem-solving
 - * RH: face recognition, attention monitoring
 - * memory: LH (words), RH (faces), objects both hemispheres
 - * attention: endogenous (LH lateralized), exogenous (bilateral)
 - split-brain findings:
 - * independent motor actions possible
 - * parallel simple tasks possible
 - * complex decisions require integration
 - possibly subcortical coordination?
 - Monteiro (2019):
 - lateralized activation (language, S1, S2) present in AgCC
 - no symmetric activation → possible role of aberrant bundles
 - lateralization alone does not explain absence of split-brain syndrome
 - Szczupak (2022):
 - new techniques reveal sparse/low-density heterotopic connections → previously underestimated
 - ~75% heterotopic connections in mouse brain
 - * mainly central CC core vs homotopic periphery
 - more numerous in malformed brains (e.g., Probst bundles)
 - * previously considered atypical
 - * may be compensatory
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Description of CC agenesis

- Glass (2008):
 - prevalence // 2/10,000
 - frequent associated central nervous system malformations (49%)
 - social impairments (often autism-like)
 - 17% chromosomal abnormalities

- risk factors: maternal age, prematurity
 - Paul (2014): overlap between ASD and AgCC, especially in communication
 - Bedeschi (2006):
 - wide variability in outcomes (from typical cognition to severe psychomotor delay)
 - partial and complete AgCC: similar frequency of associated malformations, but different types
 - * partial AgCC → posterior fossa malformations
 - * complete AgCC → cortical development malformations
 - frequent association with craniofacial abnormalities, genetic anomalies, epilepsy
 - Lau (2012):
 - deficits in social skills, imagination, theory of mind, communication, attention switching
 - no deficit in attention to detail
 - intermediate scores between controls and ASD
 - Badaruddin (2007): AgCC symptoms less severe than ASD
 - Paul (2014):
 - difficulties in attention (mainly), social functioning, thought processes, somatic complaints
 - no specific restricted interests or detail-focused attention
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Core syndrome (Brown, 2019)

(1) Reduced interhemispheric transfer of sensory-motor information

- preserved integration for simple stimuli, impaired for complex tasks
 - deficit in bimanual coordination (Mueller, 2009; Gooijers, 2013)
 - CC develops slowly + similar performance in children and AgCC → CC critical for bimanual coordination
 - performance in complex asymmetric tasks improves with age (independent of simple motor skills)
 - * interhemispheric integration especially important for complex tasks
 - deficits in tactile-spatial processing (Dunn, 2000)
 - reduced visual and auditory memory (Paul, 2016)
 - difficulty organizing new information for encoding (possible LH/RH integration issue)
 - may also reflect general resource limitation (see point 3)
 - difficulty encoding and retrieving new information (Erickson, 2014)
 - deficits in both initial learning and delayed recall
 - HERA model:
 - * right prefrontal cortex → retrieval
 - * left prefrontal cortex → encoding
 - commissurotomy: encoding and retrieval possible within each hemisphere
 - * weaker memory may reflect reduced integration
 - interpretation:
 - * verbal memory deficits due to lack of integration between LH (serial/verbal) and RH (semantic/contextual) processing
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(2) Reduced cognitive processing speed

- cognitive control differences mainly due to reduced processing speed (Marco, 2012)
 - deficits increase with task complexity (Mathias, 2004)
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(3) Deficits in complex reasoning and novel problem-solving

- interhemispheric resources required for complex cognition (Koivisto, 2000)
 - impairments mainly in complex tasks (Schieffer, 2000)
 - abstract reasoning (Brown, 2000)
 - concept formation (Fischer, 1992)
 - strategy formulation (Brown, 2012)
 - generalization (Solursh, 1965)
 - no deficits in overlearned tasks (Erickson, 2013)
 - reduced creativity: coherent but low diversity outputs (Brown, 2024)
 - creativity model: interhemispheric inhibition → diversity, then integration → coherence
 - difficulty integrating contextual information and generating hypotheses (Su, 2023)
 - difficulty generating meaningful associations between words (Turk, 2010)
 - * possibly due to reduced LH/RH integration
 - * link with memory deficits (Paul, 2016)
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(4) Difficulty integrating complex information → social difficulties (hypothesis)

- difficulties in emotional processing:
 - difficulty identifying/describing emotions (Paul, 2021)
 - * but preserved emotional experience
 - * → deficit in cognitive processing of emotion
 - impaired recognition of complex emotions (fear, anger) (Bridgman, 2014)
 - difficulty expressing emotions verbally (Pazienza, 2011)
 - reduced attention to eyes
 - impaired face recognition (complex stimuli) (Paul, 2016)
 - preserved basic emotion understanding but difficulty integrating in complex social contexts (Anderson, 2017)
 - difficulty reasoning about emotions in context
 - difficulty integrating cognitive and emotional information + anticipating consequences (Young, 2019)
 - difficulty inferring others' mental/emotional states (Kang, 2009)
 - impaired judgment of social norms (Brown, 2021)
 - deficits in emotional prosody and non-literal language (Paul, 2003)
 - * literal language preserved
 - * poor proverb/non-literal comprehension (Rehmel, 2016)

- * emotional prosody requires both hemispheres
 - * reduced integration of RH (emotional/holistic) and LH (analytic) processing
 - * deficits stronger in complex tasks
 - difficulties in behavior:
 - deficits in adaptive behavior, especially social skills (Haisley, 2025; Miller, 2024)
 - rigid application of social norms (lack of contextual adaptation) (Brown, 2021)
 - reduced flexibility (Marco, 2012)
 - deficits in planning, organization, cognitive flexibility (Mangum, 2021)
 - difficulty forming strategies, inconsistent decision-making, poor use of available information (Brown, 2012)
 - under uncertainty: increased focus on potential loss (Brown, 2012)
 - increased impulsivity / reduced inhibition
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(4) Other

- reduced self-awareness / metacognition (Mangum, 2018/2021; Kaplan, 2012)
 - Lau (2012): parents report more severe deficits than self-reports (similar to ASD)
- Schieffer (2000): difficulties mainly observed in older children
- deficits in sustained attention / inhibition (Brown, 2020)
 - mainly in adolescents
 - * possible compensation in adults?
 - * or task-related effects due to processing speed differences? [UNCLEAR]
 - difficulty maintaining conversation, but not due to lack of interest (Paul, 2014)

Cognitive science in general

Discourse theory (Mason, 2006)

- dynamic recruitment of cortical networks depending on text constraints
- previous discourse theories:
 - coarse coding
 - construction-integration framework
 - salience hypothesis
- three main hypotheses about the role of the right hemisphere (RH):
 - spillover: recruitment of RH networks when left hemisphere (LH) capacity is exceeded
 - RH involvement in inference integration
 - coarse coding theory:
 - * inferences arise from activation of broadly coded semantic information in RH
- parallel networks of discourse:
 1. Basic LH sentence network (not specific to discourse)
 - includes multiple LH regions
 - functions: syntactic processing, word meaning, visual/graphemic processing, phonological and lexical-semantic processing
 2. Coarse semantic processing network (right middle and superior temporal regions)

- activates broad semantic representations and possible inferences
 - detects overlap between activated meanings
 - 3. Coherence monitoring network (bilateral dorsolateral prefrontal cortex)
 - increased activity when coherence is low or semantic field is unusual
 - supports inference generation to restore coherence
 - 4. Text integration network (left inferior frontal + left anterior temporal regions)
 - constructs a situation model / mental representation of the text
 - 5. Perspective-taking network (bilateral medial frontal + right posterior temporal/parietal regions)
 - processes agent/protagonist perspective
 - supports understanding of intentions, motivations, emotion-related memory
 - 6. Spatial imagery network (left-dominant, bilateral intraparietal sulcus)
 - activated during high-imagery sentences or metaphors
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Ideas for relevant analyses

- compare deficits across categories (ASD vs AgCC)
 - motor deficits may differ significantly
 - compare performance across psychological tests
 - genetics
 - behavioral measures
- hypotheses about differences between ASD and AgCC
 - AgCC: primary deficit = interhemispheric disconnection
 - * impacts motor coordination, integration
 - ASD: primary deficit = local hyperconnectivity / global hypoconnectivity
 - * local/global processing bias
 - * hypersensitivity
 - possible link with attention
 - hypothesis to test : similar mechanisms in AgCC?
- correlations between different deficit domains
 - Hilton (2007): correlation between sensory and social deficits
 - * present in ASD → test whether absent in AgCC?
 - social skills <-> motor skills
 - Gomes (2008): hypersensitivity correlated with attention
 - executive functions
 - Dwyer (2014): hyperfocus correlated with quality of life
- neuroimaging correlations: morphometry / fiber density / activation / behavior
 - Alexander (2007): processing speed <-> morphometry / DTI (FA, MD)
 - Hinkley (2012): resting-state functional connectivity <-> executive functions
 - Mason (2008): fronto-parietal connectivity <-> CC genu size
 - Dapretto (2005): mirror neuron system activity <-> social ability
 - Diamond (2019): social impairment <-> splenium fiber density
 - Hett (2006): anterior commissure size <-> cognition in AgCC

- Lefebvre (2014): intelligence <-> brain volume
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Language (Julio review)

- structural connectivity (diffusion MRI) between Broca's subregions and frontal cortex
 - anterior Broca: semantics
 - posterior Broca: phonology, syntax, motor coordination
 - AgCC dyspraxia:
 - disruption of salience and fronto-parietal networks
 - symptoms may disappear → functional compensation?
 - compared to split-brain: “against the patient's will”
 - functional specialization:
 - semantics → temporal regions
 - phonology → frontal regions
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Four-level model for pathological speech

- (1) Sensorimotor control (linguistic-symbolic planning)
 - regions: temporal-parietal, Broca's area
 - sound map: left ventral premotor cortex
 - (2) Speech-motor planning
 - sequential organization of movements + co-articulation
 - involves frontal cortical areas
 - inverse models: generate motor commands
 - forward models: predict consequences of motor commands
 - * impairment → difficulty initiating sounds, start-restart behavior
 - regions (dominant hemisphere):
 - * prefrontal cortex
 - * area 6
 - * supplementary motor area (SMA)
 - * posterior parietal areas (5 and 7)
 - * Broca's and Wernicke's areas
 - (3) Speech-motor programming
 - specification of muscle movements (timing, velocity) + prosody/emotional content
 - * atypical duration?
 - rapid compensatory movements via auditory feedback
 - circuits: sensorimotor cortex, lateral cerebellum, basal ganglia
 - (4) Speech execution
 - transmission of motor commands to lower-level motor systems
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- Apraxia of speech (AOS):

- lesions in cortical motor planning areas
 - * possibly left ventral premotor cortex
 - disruption of planning and/or programming (difficult to dissociate)
 - symptoms:
 - * impaired prosody
 - * slow speech rate
 - * reduced articulation accuracy
 - * syllable-by-syllable speech
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Summary of regions of interest

- structural connectivity (diffusion MRI):
 - Broca's subregions <-> frontal cortex
- dyspraxia:
 - fronto-parietal network → resting-state fMRI
- functional specialization:
 - semantics → temporal regions
 - phonology → frontal regions
- sound map:
 - left ventral premotor cortex
 - connectivity with SMA, Broca's area
- speech-motor planning:
 - dominant hemisphere network: prefrontal cortex, area 6, SMA, posterior parietal (5, 7), Broca, Wernicke
- speech-motor programming:
 - cortical-subcortical circuits: sensorimotor cortex, lateral cerebellum, basal ganglia

Summary of key aspects of ASD

- A variety of symptoms:
 - many shared with AgCC:
 - * communication and social interaction difficulties
 - * attention switching deficits / overselective attention
 - * difficulty with novel and complex tasks
 - * reduced imagination
 - more specific to ASD:
 - * restricted and repetitive behaviors
 - * hypersensitivity
 - * atypical productive prosody? hypothesis : specificity to ASD ?
 - no clearly identified single physical/anatomical cause
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Cognitive theories

- deficit in executive functions
 - may result from fronto-parietal underconnectivity
- weak central coherence: cognitive style biased toward details

- reduced global/contextual processing
 - difficulty integrating multiple sources of information
 - complexity theory:
 - deficit in integrative context processing
 - preserved associative context processing
 - leads to detail-focused processing
 - monotropism: restricted range of interests
 - attentional competition
 - hyperawareness within attentional focus, hypoawareness outside
 - hyper-systemizing theory:
 - difficulty processing high-variance systems
 - reliance on structured/systematic processing
 - requires efficient integration
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Summary in a neuro-oriented framework

- local overconnectivity / global underconnectivity
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Summary of key aspects of AgCC

- core syndrome (not independent components) (Brown, 2019):
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(A) More specific to AgCC (less frequent in ASD)

(1) Reduced interhemispheric transfer of sensory-motor information

- motor:
 - preserved integration for simple stimuli
 - impaired integration for complex tasks
 - deficits in bimanual coordination and cross-modal tasks
 - memory:
 - difficulty encoding verbal and visual information
 - difficulty retrieving newly learned information (Erickson, 2014)
 - possible interpretation:
 - * reduced ability to integrate RH semantic/contextual information with LH verbal memory
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(2) Reduced cognitive processing speed

- cognitive control differences mainly driven by reduced processing speed (Marco, 2012)
 - deficits increase with task complexity (Mathias, 2004)
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(B) Less specific to AgCC (largely shared with ASD)

(3) Deficits in complex reasoning and novel problem-solving

- complex tasks:

- impaired in AgCC (Schieffer, 2000)
 - rely on interhemispheric resources (Koivisto, 2000)
 - affected domains:
 - abstract reasoning (Brown, 2000)
 - concept formation (Fischer, 1992)
 - strategy formulation (Brown, 2012)
 - generalization (Solursh, 1965)
 - novel vs overlearned tasks:
 - no deficit in overlearned processes (Erickson, 2013)
 - interhemispheric contribution decreases with practice (Cherbuin, 2005)
 - creativity:
 - difficulty generating stories/associations (Turk, 2010)
 - outputs coherent but low diversity (Brown, 2024)
 - possible explanation: reduced LH–RH integration
 - creativity model: interhemispheric inhibition → diversity, then integration → coherence
 - reduced attention
 - especially shifting
-

(4) Hypothesis: difficulty integrating complex information → social deficits

- emotional processing:
 - preserved emotional experience and coping (Paul, 2021)
 - impaired emotional interpretation and cognitive processing:
 - * identifying emotions
 - * describing emotions
 - * integrating cognitive and emotional information
 - * reasoning in complex social contexts
 - * judging appropriate social norms
 - impaired face recognition (complex stimuli) (Paul, 2016)
 - difficulty inferring others' mental/emotional states (Kang, 2009)
- language:
 - preserved literal language processing
 - impaired emotional prosody and non-literal language (e.g., proverbs)
 - * emotional prosody requires both hemispheres
 - * reduced integration of RH (emotional/holistic) and LH (analytic) processing
 - * deficits increase with task complexity
- behavior:
 - reduced adaptive and flexible behavior
 - * rigid application of social norms (Brown, 2021)
 - * poor use of contextual information
 - decision-making:

- * difficulty using available information
 - * inconsistent strategies
 - increased impulsivity / reduced inhibition
 - reduced planning and organization
 - no lack of social interest, but difficulty sustaining interaction
-

Why ASD and AgCC share symptoms

- AgCC: absence of corpus callosum → interhemispheric disconnection
 - ASD: local hyperconnectivity → global underconnectivity
 - both involve connectivity alterations
 - → difficulty integrating information across brain regions
-

Hypothesis about shared symptoms

- mainly affect:
 - social cognition
 - complex tasks (requiring integration)
 - core mechanism:
 - impaired integration of information from multiple sources
-

Interhemispheric integration (LH–RH)

- LH: lexical, syntactic processing
 - inferior frontal gyrus (IFG)
 - superior and middle temporal gyri (STG, MTG)
 - RH: contextual and social processing
 - superior temporal sulcus (STS): prosody, social cues, intentions
 - middle temporal gyrus (MTG): coarse semantics, affective/social processing
 - temporo-parietal junction (TPJ): perspective-taking, theory of mind
 - discourse theory (Mason, 2006):
 - LH sentence network
 - RH coarse semantic processing network
 - LH text integration network
 - bilateral perspective-taking network
 - hypothesis:
 - reduced structural/functional integrity of fronto-parietal network
 - * (e.g., reduced size or connectivity)
 - fronto-parietal network supports top-down integration across modalities and time
-

Multimodal integration deficit

- functional hypotheses:

- reduced connectivity between STS and sensory regions (visual/auditory)
 - reduced connectivity between LH and RH hubs
 - structural hypotheses:
 - reduced volume in integrative regions (TPJ, STS) (Par, 2021)
 - DTI: reduced integrity of posterior CC and inferior longitudinal fasciculus
-

Complex task processing

- functional:
 - reduced connectivity in fronto-parietal network
 - * affects reasoning, strategy, flexibility
 - structural:
 - reduced integrity of anterior CC and superior longitudinal fasciculus (SLF)
 - reduced volume in fronto-parietal regions:
 - * dorsolateral prefrontal cortex (DLPFC)
 - * posterior parietal cortex (PPC)
-

Emotion / prosody / semantic integration

- functional:
 - reduced connectivity between:
 - * RH temporal regions (STS, MTG, TPJ)
 - * LH frontal regions (IFG, STG, MTG)
 - → reduced integration of emotional information into language
 - structural:
 - reduced volume / fiber integrity in RH STS/MTG and bilateral amygdala
 - reduced integrity of splenium
 - hypotheses to be extended
-

ASD-specific (non-AgCC) hypotheses

- possibly linked to local hyperconnectivity
 - connectivity deficits in four networks (Elton, 2015):
 - default mode network (DMN) → preserved in AgCC
 - dorsal attention network (DAN)
 - ventral attention network (VAN)
 - executive control network (ECN / fronto-parietal)
 - monotropism (Dwyer, 2024):
 - atypical attention regulation
 - * increased endogenous hyperfocus
 - * increased exogenous attentional capture
 - DAN and VAN overactivation → reduced flexibility in ECN
 - hypothesis: preserved in AgCC, altered in ASD
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Hypersensitivity and attention

- ASD: strong link between hypersensitivity and attention
 - possible mechanism:
 - * overselective attention and repetitive behaviors as regulation (Liss, 2006)
 - potential neural correlates:
 - connectivity in DAN/VAN
 - parietal cortex structure
 - hypothesis:
 - hypersensitivity absent or weaker in AgCC
 - weaker correlation with attention in AgCC
-

Cerebellum

- ASD:
 - smaller cerebellum
 - altered excitation/inhibition balance (reduced inhibition)
 - abnormal cortico-cerebellar connectivity (Ramos, 2019)
 - * linked to social cognition
 - AgCC:
 - not clearly affected?
 - possibly increased cerebellar activation in resting-state (Siffrédi, 2021)
 - * alternative compensatory pathway
-

Corpus callosum (ASD and partial AgCC)

- topographic organization:
 - genu (frontal), body (parietal), splenium (occipital)
 - developmental hypothesis:
 - anterior regions develop first (genu)
 - * consistent with remnant CC in partial AgCC
 - findings:
 - fronto-parietal connectivity correlated with genu size
 - social impairment negatively correlated with splenium fiber density
 - AgCC:
 - fronto-parietal underconnectivity → executive deficits
 - preserved activation within regions
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AgCC-specific hypotheses

- primary mechanism: disrupted interhemispheric communication
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Motor deficits

- impaired coordination and multimodal integration
- hypotheses:
 - reduced connectivity between homologous motor regions (M1, SMA, parietal)
 - correlation with bimanual task performance
 - ASD may show higher connectivity and different correlation patterns in complex tasks

Processing speed

- possible reduced homotopic connectivity ?
 - ASD and partial AgCC:
 - processing speed correlated with callosal integrity
 - general white matter correlates:
 - parietal and temporal tracts
 - left middle frontal gyrus (Turken, 2008)
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Lateralization (Hinkley, 2016)

- investigate with resting-state fMRI:
 - intrahemispheric vs interhemispheric connectivity